

K-12 STEM Outreach Preparation

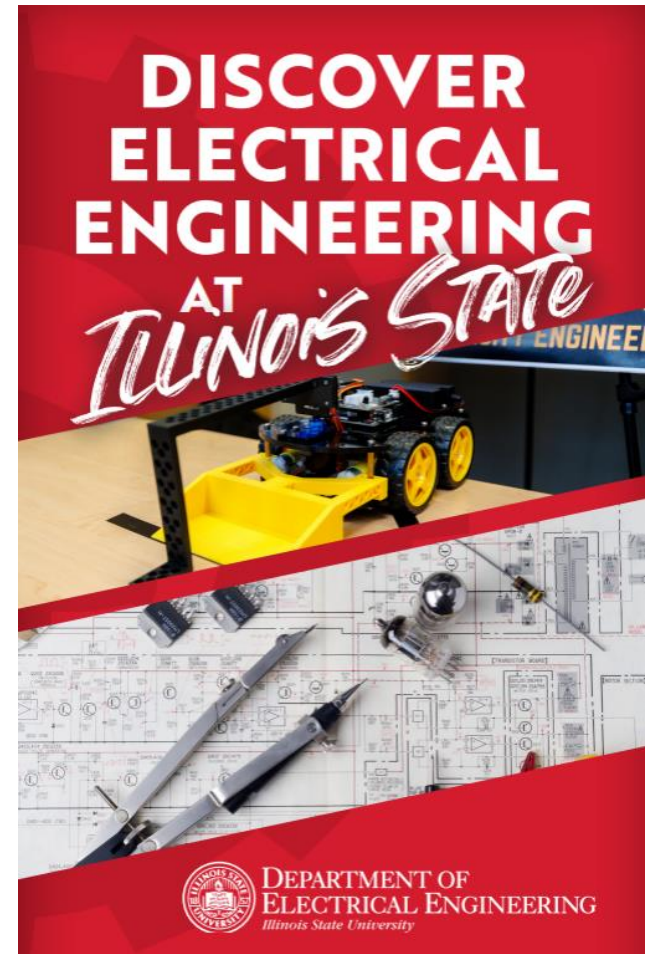
Hands-on Engineering with \$1 Kits

Preparing low-cost electrical engineering demonstrations for middle and high school visits

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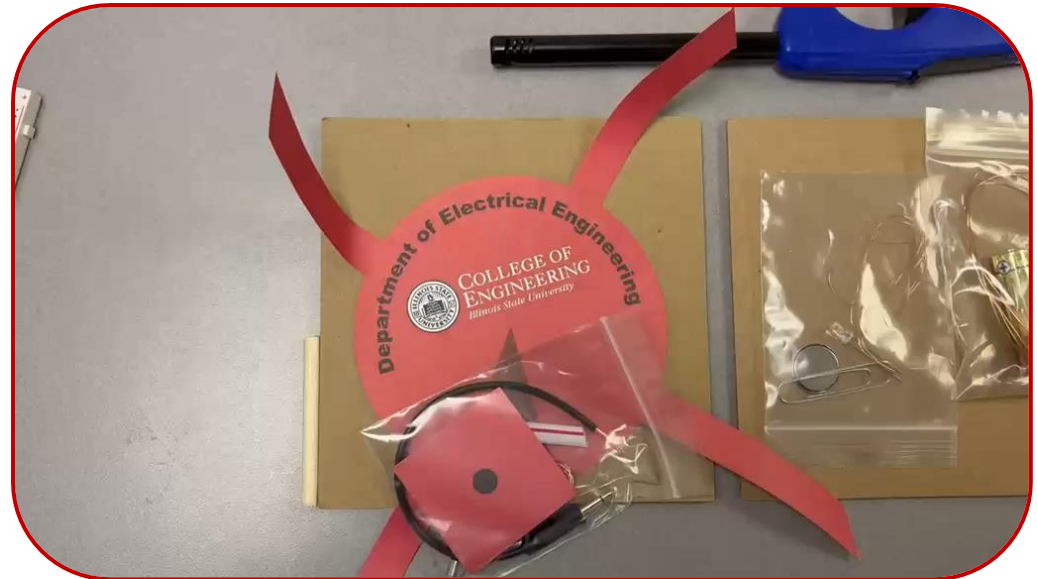


Introducing Electrical Engineering Through Hands-on STEM Outreach



The planned K–12 outreach activities introduce middle and high school students to electrical engineering through simple, low-cost, and interactive demonstrations.

Students can directly observe how engineering concepts appear in motors, sound devices, and circuit components.



Accessible

Low-cost activities
suitable for classroom
visits

Interactive

Students observe, build,
and test simple systems

Recruitment-Oriented

Builds early interest in
STEM and electrical
engineering

Goal: Make electrical engineering visible, affordable, and engaging for K–12 students.

Why This Outreach Matters

From early curiosity to engineering awareness



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Hands-on outreach helps students connect abstract engineering ideas to physical systems they can observe and test.

- **Concrete concepts** Motors, LEDs, and speakers make electricity visible.
- **Student participation** Short activities encourage students to observe and test.
- **Career awareness** Demos connect classroom science to engineering pathways.
- **Scalable outreach** Low-cost materials support repeated school visits.

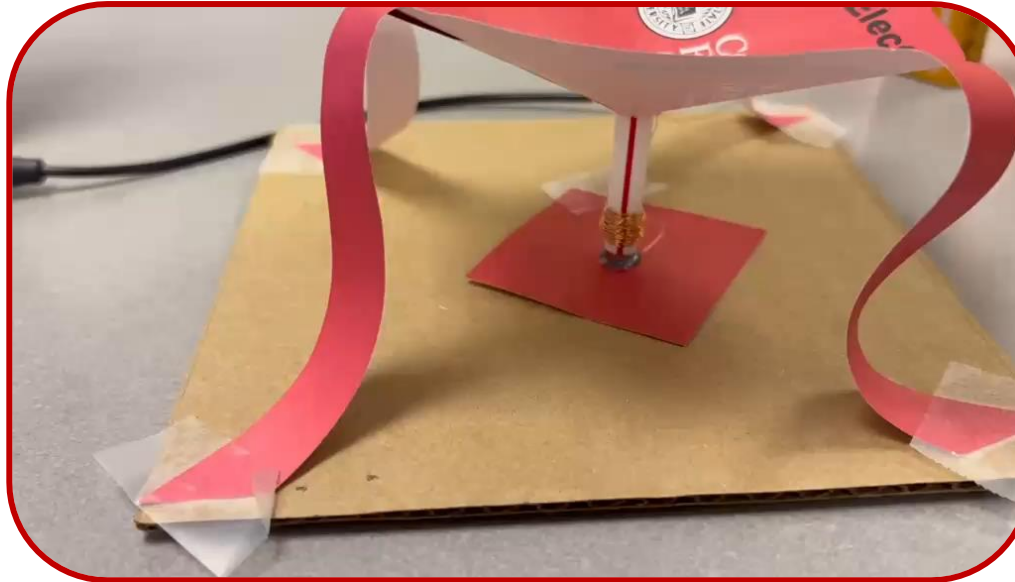


\$1 Engineering Kit Activities

Hands-on demonstrations that connect circuits to real-world behavior



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DC Motor Kit

- Demonstrates electromagnetism and energy conversion.
- Students see how current, magnetic field, and motion are connected.



Speaker/Sound Kit

- Demonstrates electrical signals, vibration, and sound.
- Connects circuits to real-world devices such as speakers and phones.



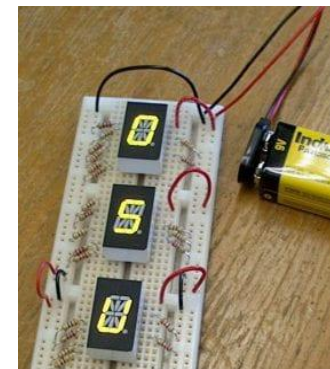
LED / Breadboard Kit

- Demonstrates circuits, polarity, resistance, and digital logic basics.
- Helps students understand how small components build larger systems.

Goal: Make electrical engineering accessible, affordable, and engaging for middle and high school students.

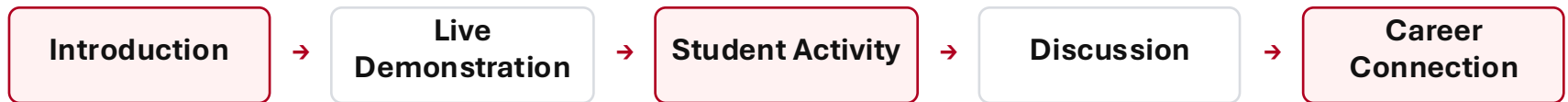
Hands-on Demonstrations Planned for School Visits

Kit Activity	Concept Demonstrated	Student Takeaway
DC Motor Kit	Current, magnetic field, and motion	Students see how electricity can create mechanical movement
Speaker/Sound Kit	Electrical signals, vibration, and sound	Students connect circuits to real-world devices such as speakers
LED/Breadboard Kit	Circuits, polarity, and resistance	Students build and test a simple working circuit
Light-Up Star Wand	Circuits, LED, and battery	Students see how electrical energy powers light



Each activity is designed to be short, visual, and suitable for classroom-based outreach.

School Visit Delivery Plan



- Each school visit will begin with a brief introduction to electrical engineering, followed by live demonstrations using the \$1 kits. Students will then participate in guided hands-on activities where they observe, build, or test simple circuits.
- The session will conclude by connecting the activities to engineering careers, college preparation, and opportunities in the College of Engineering.



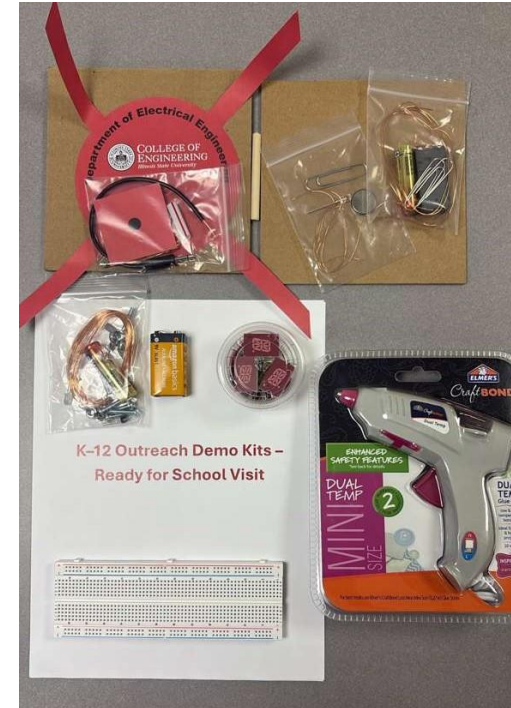
Kits organized in labeled boxes for classroom demonstrations and guided student activities.

Expected impact

- Introduces K–12 students to engineering through hands-on learning
- Supports early interest in electrical engineering and STEM
- Uses low-cost kits that scale across multiple school visits
- Strengthens College of Engineering outreach and recruitment efforts

Next steps

- Finalize instruction sheets and classroom timing
- Coordinate school visit dates and logistics
- Prepare student-volunteer support for larger visits



Prepared kits can be reused and expanded for future outreach events, open houses, and recruitment activities.

Thank you